

23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

2. UI Design Document

Group No : D13

Team Members:

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Project Title: Corporate Compliance and Certification Tracker

2.1. Functional requirements:

Role-Specific Dashboards: Different interface views tailored for HR staff, managers, and admins, so everyone only sees the tools and data they need to do their specific job.

Smart Expiry Tracking: A visual, color-coded system that instantly shows HR which employee certificates are active, expiring soon, or already expired without needing to open individual files.

Simple Data Management: Easy-to-use forms for adding new employees, assigning training programs, and linking certifications to replace messy manual spreadsheets.

Automated Reminders: A notification hub that proactively alerts the HR team about upcoming deadlines to ensure no mandatory renewals fall through the cracks.

Quick Search & Audit Reports: The ability to quickly search for specific employees, filter lists by department, and generate clean summary reports when it is time for a company audit.

2.2. Non functional requirements:

Intuitive User Experience (UX): The interface should be clean, straightforward, and easy to navigate so that HR staff can start using it immediately with minimal technical training.

Data Security: The system must securely handle sensitive employee records and compliance data behind the scenes to protect privacy and avoid legal issues.

Fast Performance: The web pages should load quickly and filter data smoothly, ensuring the system doesn't slow down even as the company grows and adds hundreds of new employee records.

Reliable Record Keeping: The UI should safely archive historical certifications, keeping the main dashboard clutter-free while ensuring past records are always a click away for reference.

2.3. List of AI tools used for building UI(if any):

(Note: While our team manually developed the core architecture, database integration, and complex JavaScript logic, AI tools were utilized to generate foundational boilerplate code, assist with basic CSS styling, and troubleshoot errors. This accelerated our workflow while allowing us to maintain full control over the final implementation.)

1. **Gemini** (Utilized to generate initial HTML structural templates, provide CSS layout ideas, and help troubleshoot UI bugs during development)
2. **GitHub Copilot** (Utilized for basic syntax auto-completion and suggesting minor bug fixes for frontend JavaScript functions)

2.4. Prompt (if any) :

Sl. No	Name of the tool	prompts
1	Gemini	"Generate a basic HTML boilerplate for an HR dashboard, including a sidebar navigation menu and a main content area."
2	Gemini	"Provide a CSS flexbox snippet to align a row of status summary cards evenly across the top of the screen."
3	Gemini	"Help me debug this JavaScript function that is throwing an error when comparing a certificate's expiry date to today's date."

2.5. List of files:

index.html (Secure login portal and authentication simulation)

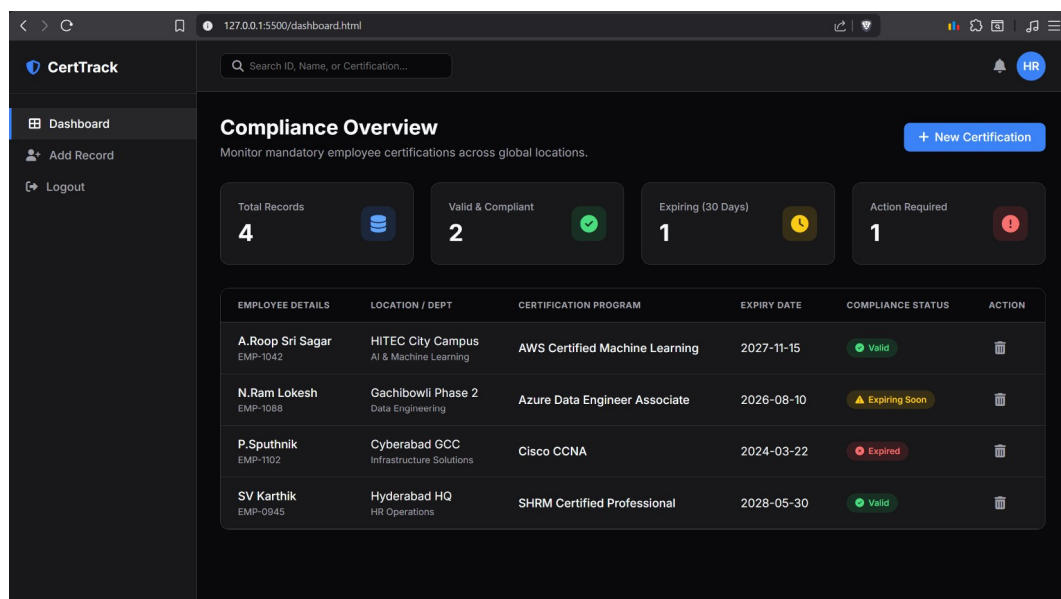
dashboard.html (Main compliance tracking view and metrics)

employee-forms.html (CRUD interface for new record insertion)

styles.css (Custom stylesheet managing Dark Mode theme, layout, and responsive UI elements)

app.js (Frontend logic managing LocalStorage, search filtering, date math, and dynamic DOM manipulation)

2.6 UI Screenshot:



2.7. GitHub Link:

<https://github.com/Venkatansu784/cert-tracker-dbms>